SPATIAL ANALYSIS OF SOIL EROSION FROM MECHANIZED FARMING IN GHANA'S RICE-GROWING REGIONS

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Abstract:

Mechanized farming is often promoted as a pathway to higher yields and greater food security, but in many parts of sub-Saharan Africa, these practices have unintended environmental consequences. In Ghana's Greater Accra Region, the K-Ricebelt Project has introduced large-scale rice production in districts such as Ningo-Prampram. This initiative employs heavy machinery and irrigation infrastructure on both flat and sloped terrain, yet few studies have examined whether these practices are contributing to soil erosion.

This research integrates geospatial data, slope analysis, and land-use overlays to assess erosion risk in mechanized farming zones. Using R and open-source GIS packages, we created digital elevation models (DEMs), slope maps, and classified slope categories (low: 0–5°, moderate: 5–12°, high: >12°). We then overlaid known mechanized farming districts and identified areas where steep slopes and mechanized activity intersect.

Preliminary results indicate that certain rice-growing districts have significant areas of steep slope under mechanization. Ningo-Prampram in particular exhibits slope classes over 12°, which, when combined with heavy tillage and reduced vegetation cover, increases susceptibility to runoff and soil loss. These findings highlight the importance of integrating environmental monitoring into agricultural expansion plans. By identifying high-risk areas, this research supports the adoption of sustainable practices such as contour ploughing, mulching, or cover crops to reduce erosion and maintain soil health.

Background Information:

- Soil erosion is a leading cause of land degradation and reduced agricultural productivity in sub-Saharan Africa.
- Mechanized farming often removes protective vegetation and exposes soil to rainfall and surface runoff.
- Prior studies in Ghana's northern regions (Salifu et al., 2021; Sodoke et al., 2024) show erosion rates as high as 50–56 t/ha/year in areas with steep slopes and poor cover management.
- Little research has been done in southern Ghana, where mechanization is expanding in regions like Ningo-Prampram under initiatives such as the K-Ricebelt Project.
- Understanding slope, rainfall patterns, and farming intensity is crucial for identifying "hotspots" of soil erosion risk.

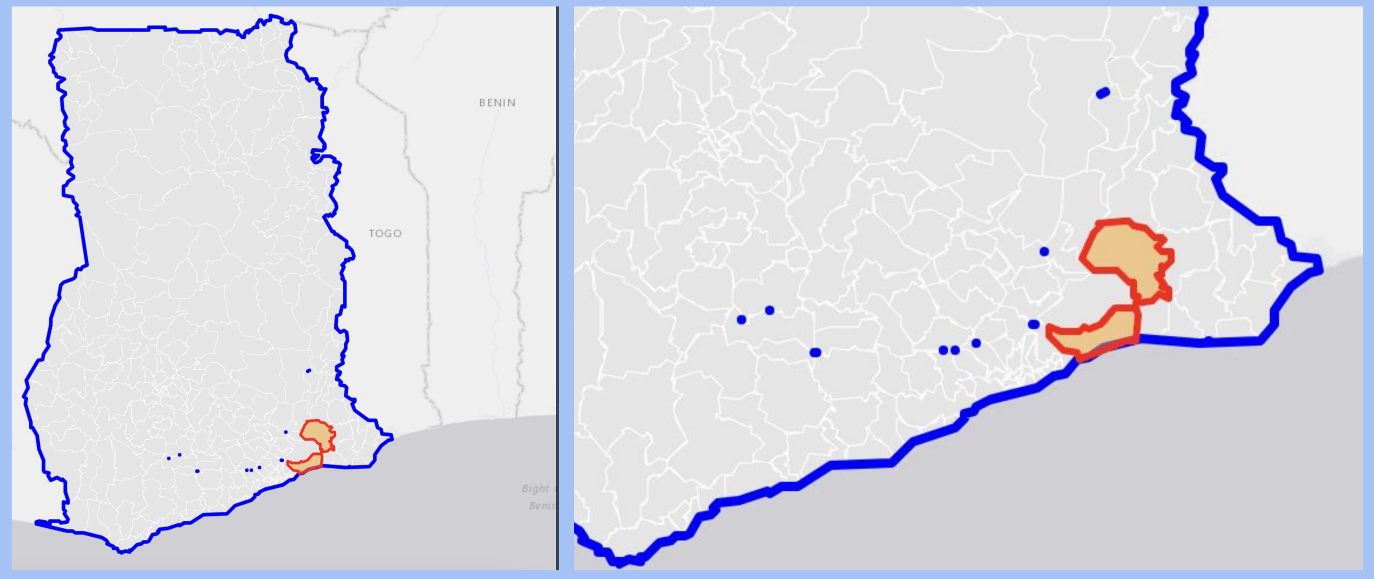

Fig. 1: Ghana boundary with highlighted mechanized districts.

Method/Materials:

- **Software and Tools:** R (sf, terra, tmap, geodata), DEM datasets (elevation_30s), shapefiles of Ghana administrative districts.
- **Workflow:**
  - Download and preprocess elevation data (DEM) for Ghana.
  - Clip DEMs to mechanized farming districts (e.g., Ningo-Prampram, Pru, North Tongu).
  - Derive slope rasters and classify slope into three categories: low, moderate, high.
  - Overlay district boundaries and highlight areas of mechanized farming.
  - Create maps and histograms for elevation and slope distributions.
- **Validation**: Visual inspection and comparison with known agricultural regions; literature review to cross-check mechanized farming sites.

Results:

- **Slope Classes:**
  - Ningo-Prampram district shows notable portions of land with slopes >12°.
  - Pru and North Tongu also contain moderate-slope areas under mechanization.
- **Maps Produced:**
  - Ghana boundary with highlighted mechanized districts.
  - DEM showing elevation variations and histograms of pixel elevation distribution.
  - Slope maps showing low (0–5°), moderate (5–12°), and high (>12°) slope classes.
- **Hotspot Detection:**
  - Areas with high slope (>12°) and known mechanized activity are flagged as high-risk erosion zones.

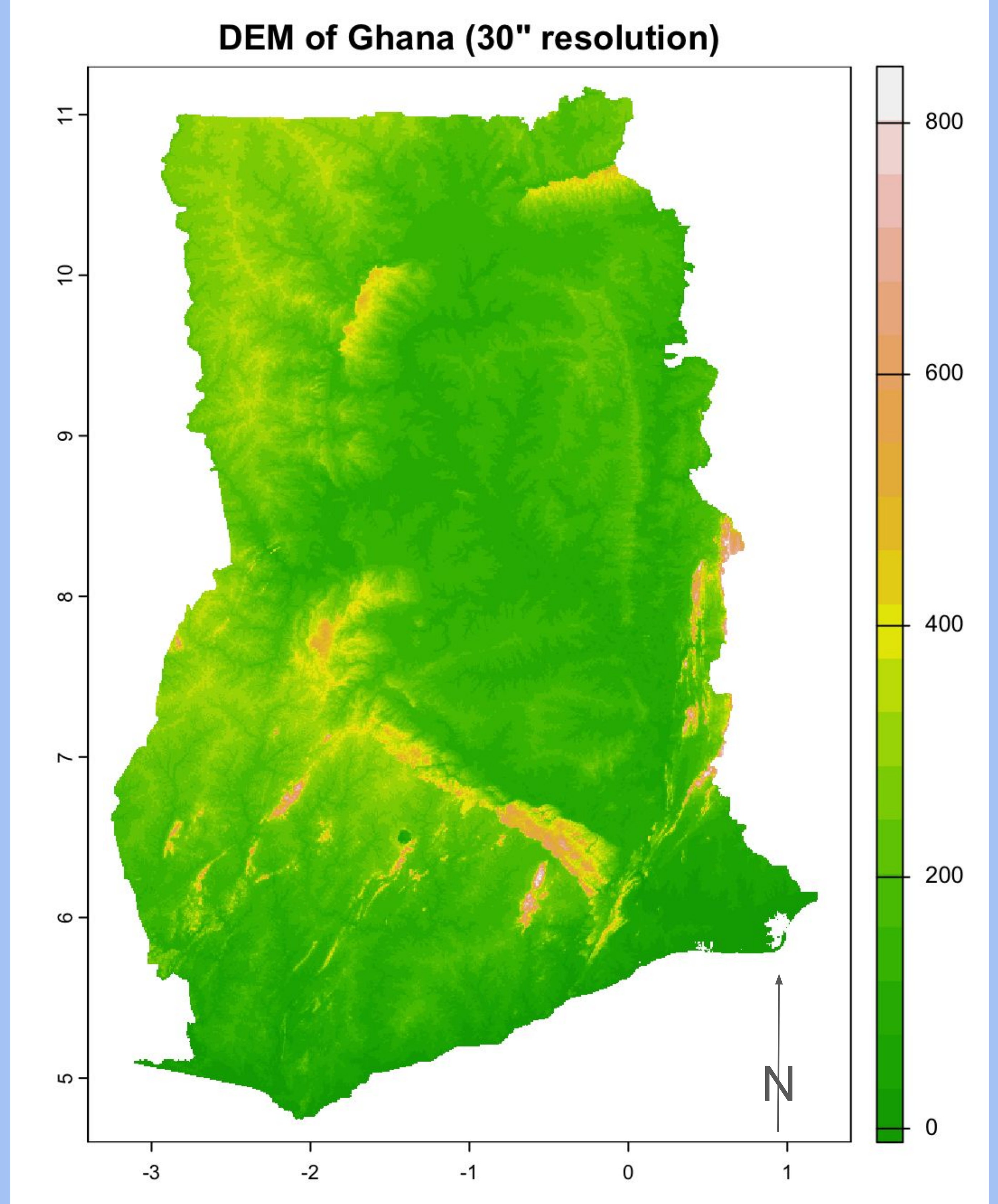

Fig. 2: DEM showing elevation variations.

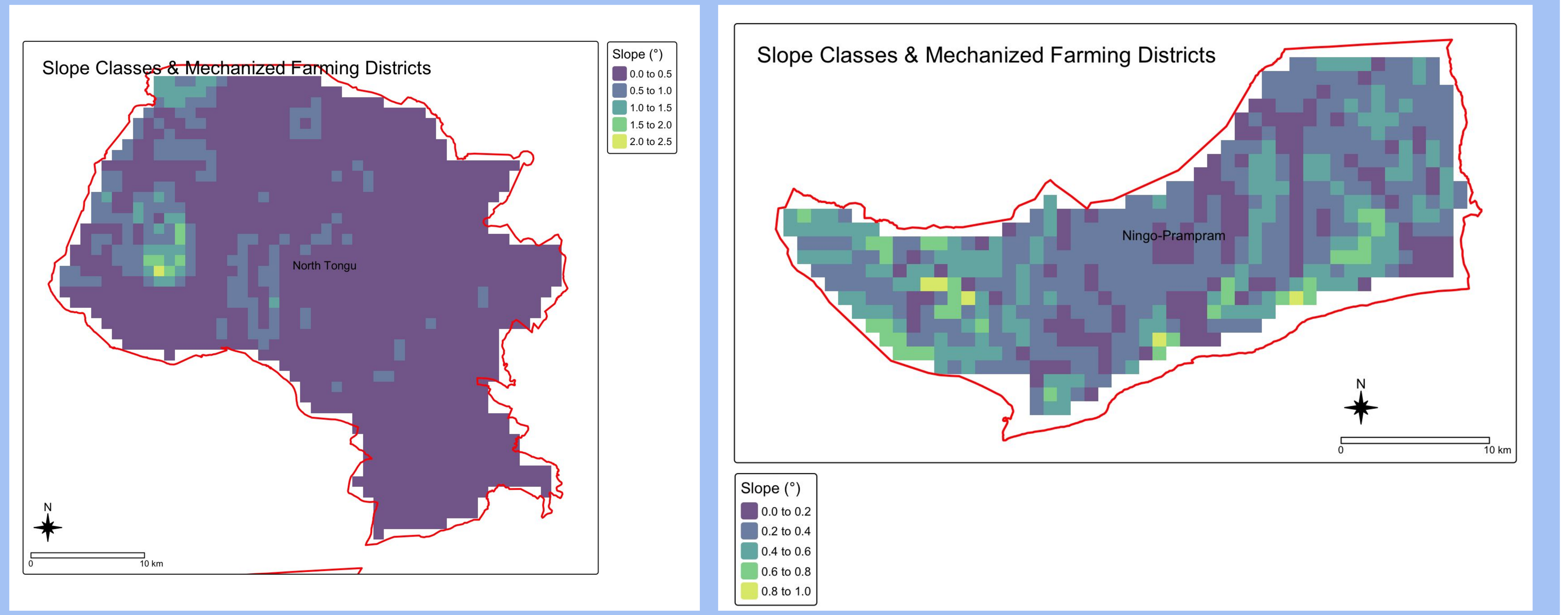

Fig. 3: Slope classification map with hotspots

Discussion/Conclusion:

- Mechanized farming in slope-intense districts increases erosion vulnerability, even in regions with relatively low tectonic hazard.
- Erosion hotspots correlate strongly with slope steepness and lack of vegetation cover rather than rainfall variability alone.
- Sustainable land management practices are needed in these areas to mitigate soil loss and preserve agricultural productivity.
- Future research will integrate rainfall data and explore the potential of real-time monitoring using satellite imagery.
- **Long-term goal**: develop guidelines for farmers and planners on where mechanization should be coupled with conservation practices.

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Credit Authorship Contribution Statement:

Research and poster preparation were completed by Helena Kim under the guidance of Prof. Marsellos, who provided supervision and feedback during the project's development. In addition, a peer-reviewed paper was prepared and submitted at the Geoscience and Remote Sensing Magazine.

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